24PY901

Quantum Principles, Structures and Computing

Module 2 – Unit 2

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Quantum Computation using Qiskit

Setting up a virtual environment

References: Qiskit documentation

Introduction

This document provides step-by-step instructions to create a reproducible Conda environment for Qiskit along with a scientific Python stack.

Step 1: Create YAML File

Create a file named `qiskit-env.yml` with the following content:

```
name: qiskit-env
channels:
- conda-forge
- defaults
dependencies:
- python=3.10
- numpy
- scipy
- matplotlib
- jupyterlab
- ipykernel
- pip
- pip:
- qiskit
- qiskit-aer
- qiskit-ibm-runtime
```

Step 2: Create the virtual environment

Open Anaconda Prompt (Miniconda) and run:

```
conda env create -f qiskit-env.yml
```

Step 3: Activate the environment

```
conda activate qiskit-env
```

Step 4: Launch JupyterLab

```
jupyter lab
```

Step 5: Test the Setup

Create a notebook and run:

```
from qiskit import QuantumCircuit

qc = QuantumCircuit(2)
qc.h(0)
qc.cx(0, 1)
qc.draw('mpl')
```

If a circuit diagram appears, the setup is successful.

Step 6: Deactivate the environment

After the work is over, deactivate the environment.

```
conda deactivate
```

This will revert to the base environment.

Conclusion

This setup provides a stable and reproducible environment suitable for teaching, research, and experimentation in quantum computing.